Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph: A

/ \

B C

/ \ / \

D E F G

Task: Starting from node A, perform a Breadth-First Search (BFS). Complete the following table.

Iteration Queue Visited Node

1

2

3

4

5

6

7

Questions:

1. Write the BFS traversal order.

2. Show the queue contents after each iteration.

SOLUTION:

```
     A
    /  \
   B    C
  / \  / \
 D   E F   G
```

BFS Dry Run Table

Iteration	Queue (after visiting node)	Visited Node
1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	Empty	G

Answer 1: BFS Traversal Order

A → B → C → D → E → F → G

Answer 2: Queue Contents After Each Iteration

Iteration	Queue
1	B, C
2	C, D, E
3	D, E, F, G
4	E, F, G
5	F, G
6	G
7	Empty

2. Depth-First Search (DFS) Using the same graph,

A

/ \

B C

/ \ / \

D E F G

Perform Depth-First Search (DFS) starting from A.

Complete the table.

SOLUTION:

```
     A
    /  \
   B    C
  / \  / \
 D   E F   G
```

DFS Dry Run Table (Left to Right)

Iteration	**Stack (Top → Right)**	**Visited Node**
1	C, B	A
2	C, E, D	B
3	C, E	D
4	C	E
5	G, F	C
6	G	F
7	Empty	G

Answer 1: DFS Traversal Order

A → B → D → E → C → F → G

Answer 2: Stack Contents After Each Iteration

Iteration	**Stack**
1	C, B
2	C, E, D
3	C, E
4	C
5	G, F
6	G
7	Empty

3. Initial State :

1 3 6

5 0 2

4 7 8

Goal State:

1 2 3

4 5 6

7 8 0

Task

Trace Breadth-First Search (BFS) until the goal state is reached.

Fill in the table.

Iteration Current State Queue Before Expansion Queue After Expansion

1

2

3

4

5

Questions

1. Which node is expanded in each iteration?

2. How many states are generated in total?

3. Write the complete solution path from the initial state to the goal state.

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

SOLUTION:

Initial State

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

7 8 0

Dry Run Table (First 5 Iterations)

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	Initial State	Initial	4 successor states
2	Successor 1	4 states	6–7 states
3	Successor 2	Remaining states	More successors added
4	Successor 3	Remaining states	More successors added
5	Successor 4	Remaining states	More successors added

Answers

1. Which node is expanded in each iteration?

- Iteration 1: Initial State
- Iteration 2: First generated state
- Iteration 3: Second generated state
- Iteration 4: Third generated state
- Iteration 5: Fourth generated state

2. How many states are generated in total?

- The goal is **unreachable**, so BFS continues exploring all reachable states (181,440) before stopping.

3. Write the complete solution path from the initial state to the goal state.

- No solution exists. There is no path from the given initial state to the goal state.

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

- BFS explores states level by level.
- Each move has the same cost (1 move).
- Therefore, the first time the goal is found is guaranteed to be the shortest path.